Survey Paper Topics -

1. AI for energy-efficient computing
2. AI in edge cloud collaborative architectures
3. Multi-Agent collaboration in embodied AI

1. AI for Energy-Efficient Computing

Scope:
Covers AI algorithms and system optimizations aimed at reducing power consumption — model compression, quantization, neuromorphic hardware, green data centers, energy-aware scheduling.

Why it's promising:

- Sustainability + AI is a **hot IEEE vertical**.
- Strong cross-disciplinary pull (AI + hardware + green tech).
- Governments and industry pushing net-zero targets → relevance to funding & citations.

Research gap:

- Many papers focus on specific optimizations; fewer surveys integrate **algorithmic, hardware, and deployment perspectives** together.

Potential IEEE fit: *IEEE Transactions on Sustainable Computing*, *IEEE Access*, *IEEE Transactions on Green Communications and Networking*.

Papers

AI for Energy-Efficient Computing

- A 2022 survey focused on **optimization models for energy-efficient computing systems**, particularly in CMOS-based architectures, scheduling, and power-aware ICT systems MDPI.
- A broader **2024 open-access survey** addresses green computing for massive IoT networks—covering edge, fog, and cloud paradigms, with energy-aware architecture, hardware, scheduling, and virtualization ScienceDirect.

2. AI in Edge–Cloud Collaborative Architecture

Scope:
Focuses on hybrid AI deployments that split workloads between **edge devices** (low-latency inference) and **cloud servers** (heavy model training/storage).

Why it's promising:

- Growing due to IoT, autonomous vehicles, and 5G/6G networks.
- Edge–cloud orchestration + AI model partitioning is relatively **under-surveyed compared to cloud-only AI**.
- Fits industrial IoT, telco, and smart city research agendas.

Research gap:

- Need for integrated review of **model offloading, energy trade-offs, privacy, and latency** under real-world constraints.

Potential IEEE fit: *IEEE Internet of Things Journal*, *IEEE Transactions on Cloud Computing*, *IEEE Network*.

Papers

AI in Edge–Cloud Collaborative Architecture

- **"Edge-Cloud Polarization and Collaboration: A Comprehensive Survey for AI"** – arXiv (Nov 2021)
  Reviews architectures and collaborative mechanisms between edge and cloud for AI workloads. arXiv
- **"A Survey on Collaborative DNN Inference for Edge Intelligence"** – arXiv (July 2022)
  Focuses on collaborative inference models involving cloud, edge, and end devices. arXiv
- **"A Survey on Integrated Computing, Caching, and Communication in the Cloud-to-Edge Continuum"** – *Computer Communications* (Elsevier, April 2024)
  Open-access review of AI-driven resource coordination in edge-cloud architectures. ScienceDirect

3. Multi-Agent Collaboration in Embodied AI

Scope:
Reviews systems where **multiple AI agents**, often with physical embodiments (robots, drones, vehicles), collaborate in real-world or simulated environments.

Why it's promising:

- Combines **reinforcement learning, robotics, and communication protocols**.
- Strong novelty — multi-agent + embodied setting is still **underrepresented in surveys** compared to single-agent or disembodied agents.
- Huge relevance for autonomous fleets, swarm robotics, and collaborative manufacturing.

Research gap:

- Very few comprehensive surveys that unify **perception, decision-making, coordination, and embodiment**.
- Emerging area — could set a benchmark in IEEE literature.

Potential IEEE fit: *IEEE Transactions on Robotics*, *IEEE Transactions on Cognitive and Developmental Systems*, *IEEE Access*.

Papers

Multi-Agent Collaboration in Embodied AI

- **"Multi-agent Embodied AI: Advances and Future Directions"** – arXiv (May 2025)
  A brand-new survey on embodied AI systems where multiple agents interact in real-world settings. arXiv
- **"Multi-Agent Reinforcement Learning: A Comprehensive Survey"** – arXiv (Dec 2023)
  Although focused on MARL broadly (not necessarily embodied contexts), it's a major resource for multi-agent learning frameworks. Reddit

Key Papers on Multi-Agent Collaboration in Embodied AI

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1. Starter / Textbook-Level (Basics)

- **Sutton & Barto** — *Reinforcement Learning: An Introduction (2nd Edition)*.
  Foundational textbook on RL concepts: MDPs, value functions, policy gradients.

- **Mnih et al., 2013** — *Playing Atari with Deep Reinforcement Learning (DQN)*.
  Breakthrough deep RL work using CNNs to play Atari directly from pixels.

- **Schulman et al., 2017** — *Proximal Policy Optimization (PPO)*.
  Widely used policy-gradient algorithm; stable and reliable baseline.

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2. Core Multi-Agent RL (Algorithms & Surveys)

- **Lowe et al., 2017** — *MADDPG: Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments*.
  Introduced CTDE paradigm for multi-agent actor-critic training.

- **Rashid et al., 2018** — *QMIX: Monotonic Value Function Factorisation*.
  Proposed value decomposition for cooperative MARL.

- **Yu et al., 2021** — *MAPPO/IPPO: PPO Variants for Multi-Agent Cooperation*.
  Showed PPO variants achieve strong performance in cooperative tasks.

- **Huh & Mohapatra, 2024** — *Multi-Agent Reinforcement Learning: A Comprehensive Survey*.
  Survey connecting MARL with game theory and ML foundations.

- **Amato et al., Dec-POMDP surveys** — *Decentralized POMDPs*.
  Formal framework for multi-agent sequential decision-making under uncertainty.

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3. Credit Assignment, Coordination & Communication

- **Foerster et al., 2018** — *COMA: Counterfactual Multi-Agent Policy Gradients*.
  Introduced counterfactual baseline for cooperative MARL credit assignment.

- **Zhou et al., 2020** — *Learning Implicit Credit Assignment (LICA)*.
  Actor-critic framework for implicit credit assignment.
- **Li et al., 2022** — *Deconfounded Value Decomposition*.
  Improves robustness in cooperative MARL by addressing confounding factors.

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4. Embodied AI & Benchmarks

- **Savva et al., 2019** — *Habitat: A Platform for Embodied AI Research*.
  Photorealistic simulation environment for navigation tasks.
- **Xia et al., 2019** — *iGibson: Interactive Gibson Environment*.
  Interactive physics-rich environment for manipulation and navigation.
- **Mu et al., 2021** — *ManiSkill: A Learning-from-Demonstration Benchmark*.
  Benchmark for robotic manipulation using physics-based simulation.
- **Feng et al., 2025** — *Multi-Agent Embodied AI: Advances and Future Directions*.
  Survey focusing on multi-agent collaboration in embodied AI.

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5. LLMs & Robotics (Recent Trends)

- **Ahn et al., 2022** — *SayCan: Grounding Language in Robotic Affordances*.
  Used LLMs (PaLM) to sequence robot skills via affordances.
- **Driess et al., 2023** — *PaLM-E: An Embodied Multimodal Language Model*.
  Large multimodal LLM designed for embodied reasoning.
- **Brohan et al., 2022/2023** — *RT-1 and RT-1-X: Robotics Transformer*.
  Transformer-based models trained on large real-robot datasets.
- **Huang et al., 2022** — *Inner Monologue: Embodied Reasoning with LLM Planning*.
  Closed-loop embodied reasoning and planning with LLMs.

📘 Key Concepts in Multi-Agent Collaboration & Embodied AI

🔹 Reinforcement Learning (RL)

- **Markov Decision Process (MDP)** — Formal model with states, actions, transitions, rewards, discount factor.
- **Value Function** — Estimates long-term return from a state or state-action pair.
- **Policy** — Mapping from states to actions. Can be deterministic or probabilistic.
- **Q-learning** — Value-based RL that learns action-value function.
- **Policy Gradient** — Optimizes policies directly using gradient ascent on expected return.
- **Actor-Critic** — Combines policy learning (actor) with value learning (critic).
- **Exploration vs Exploitation** — Balance between trying new actions vs using known good ones.

🔹 Multi-Agent Reinforcement Learning (MARL)

- **Multi-Agent System (MAS)** — Multiple agents interact in a shared environment, each with partial info.
- **Centralized Training, Decentralized Execution (CTDE)** — Train agents with global info, but deploy with local info.
- **Dec-POMDP** — Decentralized partially observable Markov decision process; formal model for MARL tasks.
- **Joint Action Space** — Combined action space of all agents (grows exponentially with #agents).
- **Partial Observability** — Each agent sees only part of the environment.
- **Non-Stationarity** — Environment changes as other agents learn simultaneously.
- **Credit Assignment Problem** — Determining each agent’s contribution to team reward.
- **Coordination Graphs** — Factorization of joint value function into local agent interactions.
- **Emergent Communication** — Agents developing protocols/signals to coordinate actions.

🔹 Core MARL Algorithms

- **MADDPG** — Multi-agent actor-critic with centralized critic, decentralized actors.
- **QMIX** — Value decomposition for cooperative tasks (monotonic mixing of agent utilities).
- **MAPPO** — Multi-agent PPO (strong baseline for cooperative environments).

- **COMA** — Counterfactual Multi-Agent Policy Gradient, solves credit assignment via counterfactual baseline.

🔹 Hierarchical Learning

- **Low-Level Skills** — Primitive policies (walk, grasp, turn).
- **High-Level Policy** — Chooses and sequences low-level skills to achieve tasks.
- **Option Framework** — Temporal abstraction in RL; policies over extended actions.

🔹 Imitation Learning

- **Behavior Cloning (BC)** — Supervised learning on expert demonstrations (state → action).
- **Inverse Reinforcement Learning (IRL)** — Learn reward function from demonstrations.
- **Generative Adversarial Imitation Learning (GAIL)** — GAN-like framework to imitate expert behavior.

🔹 Game-Theoretic Concepts in MARL

- **Nash Equilibrium (NE)** — No agent can improve its reward by unilaterally deviating.
- **Pareto Efficiency** — No agent can be made better off without making another worse off.
- **Correlated Equilibrium (CE)** — Equilibrium where agents may coordinate via shared signals.
- **Best Response Dynamics** — Iterative improvement by responding optimally to others' strategies.
- **No-Regret Learning** — Agents learn strategies ensuring regret (loss vs best fixed policy) goes to zero.

🔹 Embodied AI

- **Embodiment** — AI agents have physical presence (robots, drones, vehicles).
- **Perception-Action Loop** — Continuous cycle: perceive environment → plan → act → new perception.
- **Interactivity** — Agents actively interact with and modify their environment.
- **Continual Learning** — Agents improve over time via experience and adaptation.

🔹 Benchmarks & Simulators

- **Habitat** — High-speed simulator for embodied navigation.
- **iGibson** — Interactive simulation for navigation and manipulation.
- **ManiSkill** — Benchmark for robot manipulation skills.
- **SMAC (StarCraft Multi-Agent Challenge)** — Popular MARL benchmark for cooperation.

🔹 Generative Models & LLMs for Embodied AI

- **LLMs (Large Language Models)** — Enable reasoning, communication, and planning in natural language.
- **SayCan** — Uses LLMs to ground robot actions in affordances.
- **PaLM-E** — Multimodal LLM adapted for embodied reasoning.
- **RT-1 / RT-1-X** — Transformer models trained on large-scale real robot data.
- **Inner Monologue** — Uses LLMs for closed-loop embodied planning with feedback.

🔹 Challenges & Open Problems

- **Scalability** — Exponential growth of joint action/state space with more agents.
- **Sim2Real Transfer** — Moving policies from simulation to real-world robots reliably.
- **Safety & Trust** — Ensuring robust and safe collaboration.
- **Human-AI Collaboration** — Natural communication, role allocation, alignment with human values.
- **Social Norms** — Embedding fairness, roles, and conventions into multi-agent cooperation.